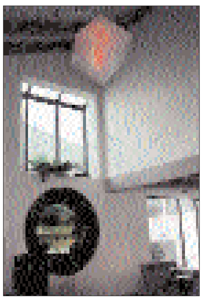
can be used as the audio source. The circuits in the Projector are thus able to translate this audio into laser images that are projected out of the Projector and into the Imaging Cube. Thus, by modulating audio instead of the laser light itself, one can control the image that is formed.

Because of the arrangement of the optical imaging material within the Cube, the projected laser images appear like dancing holograms. One can walk around and view the results from any angle.

A second method of displaying holograms entails using high-power lasers and a larger Imaging Cube suspended overhead. In this arrangement, the Projector unit containing the lasers and control systems is remotely mounted, and projects from a distance into the Cube. The projected images materialise with no apparent origin because laser beams are not visible in the air until they hit the Cube's imaging material. So there's no way to tell where the images are coming from!



Images projected into the LaserCube appear to float in space

An Example Application

There are several uses for holography besides just entertainment: advertising and medical applications are just two of them. But why is true 3D so difficult? The answer is basically the computational power required. As an example, Apple proudly mentions Voxel Inc.'s Digital Holography System (DHS), which uses Apple computers: "A patient with a brain tumour lies motionless on the operating table. The surgeon must effectively remove the tumour without damaging surrounding tissue. Previously, this was undertaken without the ability to capture in 3D the anatomical relationships within the patient's body. But now, Voxel's Mac-based DHS gives surgeons a life-size, 3D X-ray view of the patient.

"Ordinary X-rays only provide 2D representations, and technologies such as MRI and CT provide only 'slices', each representing a 2D picture of a single plane within the body—the surgeon and radiologist are left to imagine a meaningful whole. Surgeons have long needed a 3D X-ray view of the patient, giving them a clear picture, for example, of how deeply to cut to completely remove a tumour. By providing a life-size, transparent 'twin' of the patient, Voxel's technology does just that.

"If ever there was an application that requires extreme processing power, it's volume imaging," says Voxel founder and CIO Michael Dalton.

So how are 2D 'slices' turned into holograms? Surgical candidates first undergo a CT or MR scan, resulting in a series of 2D image cross-sections that are converted to DICOM (Digital Image Communications in Medicine) format. The DICOM images are transferred to a computer. A piece of software called the Voxpad then reconstructs the 2D slices into a 3D view of the anatomy, which simulates the corresponding 'Voxgram'.

"We think the two initial markets (for holo-TV) will be in medical visualisation and military applications"

Dr Harold "Skip" Garner, Jr
Professor of Biochemistry and Internal Medicine, University of Texas Southwestern Medical Center at Dallas

In doing so, the computer must process from 20 to 120 megabytes of pixel data as many times per second as possible to achieve the real-time 3D manipulation required to approximate the Voxgram. "The raw processing power of this application really stresses computer systems," says Dalton.

Voxel's DHS uses 2D slices. That's the key here. It's no simple matter to display holographic objects in motion—to just use multiple cameras to record a scene, then play it back in 3D. But steps have been taken along that direction too.

2020: Holo-TV?

This year, as recently as June, the first 'true' 3D movies were created, by Dr Harold "Skip" Garner, professor of biochemistry and internal medicine at UT Southwestern, at the UT Southwestern Medical Center. It's been reported: "In a small research laboratory at UT Southwestern Medical Center, a grainy, red movie of circling fighter jets emerges from a table-top black box, while nearby, a video of a rotating human heart hangs suspended in a tank of gooey gel."

The report in *Science Daily* goes on to say that such movies will not be commercially viable for entertainment purposes anytime soon, but Dr Garner and his holo-TV have been featured in *Popular Science*'s June 2005 issue. The magazine put Dr Garner's achievement in a list of the top five "great ideas for the future."

"An important next step is to take our proof of principle technology that we have now, and move it into a commercial entity," said Dr Garner. "We think the two initial markets will be in medical visualisation and military applications, such as heads-up displays for helmets and military aircraft and coordinating battlefield information."

"I predict that by the year 2020, that being the year of 'perfect vision,' we will have holo-TV in our homes," said Dr. Michael Huebschman, a researcher in Dr Garner's lab and one of the developers of the technology.

Dr Garner's system is based on a small chip covered with about a million tiny mirrors. According to *Popular Science*, the basic principle is this: a digital micromirror device (DMD) is used, which is made up of nearly a million reflective panels. Each of these reflective panels can be angled by a computer several thousand times a second (!) to reflect or deflect beams of light, thus producing moving pictures. The DMD can be programmed to produce a desired image.

Dr Garner's key insight was that he could use laser light on the DMD instead of using a regular projection bulb. He programmed the DMD to reflect a sequence of 2D interference patterns—called interferograms—that alter the laser in such a way that it produces a 3D hologram. (The summation of two or more interacting electromagnetic waves—light, for example—is commonly known as interference between the waves). So what's happening here is basically a sequence of 2D interferograms becoming responsible for a 3D image.